Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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OBSERVATÓRIO NACIONAL
MINISTÉRIO DA CIÊNCIA E TECNOLOGIA
PROGRAMA DE PÓS-GRADUAÇÃO EM ASTRONOMIA

Espectroscopia no infravermelho de estrelas M

Diogo Martins Souto

Orientadora:

Dra. Kátia Cunha

Início do Doutorado: Setembro de 2012

Bolsa: Capes de 09/2012 a 08/2016

Rio de Janeiro
Março de 2015

1 Projeto

2 Situação Acadêmica

– Disciplinas cursadas

Ano/Semestre	Disciplina	Conceito	Créditos
2013/1	Nucleosíntese e evolução química na era dos grandes surveys	A	4
2013/1	Estrutura e dinâmica da Galáxia	A	4

– Eventos e outras atividades

- Participação na escola; First Astrobiology school at Observatório Nacional. 2014, realizado no Rio de Janeiro, ON-MCTI.

- Participação na escola; ALMA and the Brazilian Community Workshop. 2014, realizado no Rio de Janeiro, ON-MCTI.

- Participação da escola XIX Ciclo de Cursos Especiais ON/MCT, 2013, nos seguintes temas: *Thermal Observations of Small Bodies*, Prof. Josh Emery (University of Tennessee, United States), *Chemo-dynamical Evolution of Galaxies*, Prof. Gerhard Hensler (University of Vienna, Austria), *Deep Galaxy Surveys, Initial Phases of Galaxy Evolution*, Prof. Olivier LeFèvre (Laboratoire d'Astrophysique de Marseille, France), *Relativistic Cosmology of the Accelerating Universe* Prof. Roy Maartens (University of the Western Cape, UK/South-Africa).

- Participação na XXXVIII REUNIÃO ANUAL DA SAB. Poster apresentado: Metalicidades de estrelas gigantes vermelhas do aglomerado aberto NGC 2420. 2014. Autores: Diogo Souto, Katia Cunha e Verne Smith.

– Artigos Publicados e/ou Sumbetidos

Katia Cunha, et al., *Sodium and Oxygen Abundances in the Open Cluster NGC 6791 from APOGEE H-band Spectroscopy*, Apjl, 2015. Status, publicado.

Shetrone, M et al., *The APOGEE Spectral Line List for H band Spectroscopy*, 2015. Status, submetido à ApjS.

– Exame de Qualificação

Realizado e aprovado com média final 7.1, tema: Abundâncias estelares em Não-ETL.

– Exame de Proficiência

Realizado no final de 2013.

3 Situação do projeto à época do último relatório

4 Descrição detalhada da pesquisa

5 Próximas etapas

6 Bibliografia